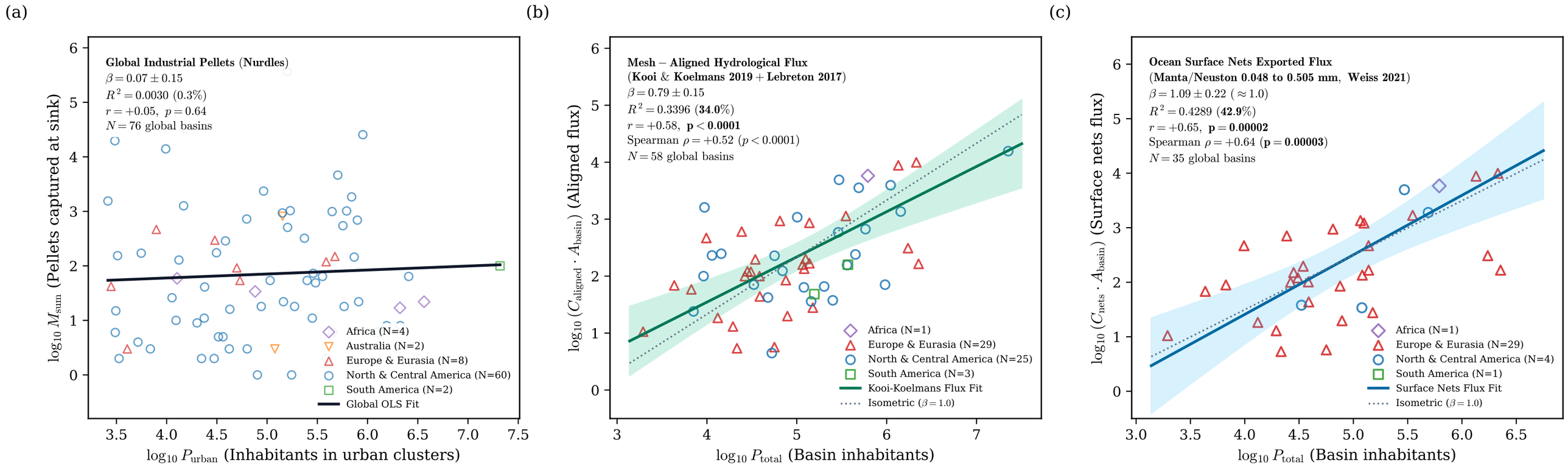

(a)
Global Industrial Pellets (Nurdles)
$\beta = 0.07 \pm 0.15$
$R^2 = 0.0030$ (0.3%)
$r = +0.05$, $p = 0.64$
$N = 76$ global basins
Africa (N=4)
Australia (N=2)
Europe & Eurasia (N=8)
North & Central America (N=60)
South America (N=2)
Global OLS Fit
$\log_{10} M_{\text{sum}}$ (Pellets captured at sink)
$\log_{10} P_{\text{urban}}$ (Inhabitants in urban clusters)
(b)
Mesh – Aligned Hydrological Flux
(Kooi & Koelmans 2019 + Lebreton 2017)
$\beta = 0.79 \pm 0.15$
$R^2 = 0.3396$ (34.0%)
$r = +0.58$, $\mathbf{p < 0.0001}$
Spearman $\rho = +0.52$ ($p < 0.0001$)
$N = 58$ global basins
Africa (N=1)
Europe & Eurasia (N=29)
North & Central America (N=25)
South America (N=3)
Kooi-Koelmans Flux Fit
Isometric ($\beta = 1.0$)
$\log_{10}(C_{\text{aligned}} \cdot A_{\text{basin}})$ (Aligned flux)
$\log_{10} P_{\text{total}}$ (Basin inhabitants)
(c)
Ocean Surface Nets Exported Flux
(Manta/Neuston 0.048 to 0.505 mm, Weiss 2021)
$\beta = 1.09 \pm 0.22$ (≈ 1.0)
$R^2 = 0.4289$ (42.9%)
$r = +0.65$, $\mathbf{p = 0.00002}$
Spearman $\rho = +0.64$ ($\mathbf{p = 0.00003}$)
$N = 35$ global basins
Africa (N=1)
Europe & Eurasia (N=29)
North & Central America (N=4)
South America (N=1)
Surface Nets Flux Fit
Isometric ($\beta = 1.0$)
$\log_{10}(C_{\text{nets}} \cdot A_{\text{basin}})$ (Surface nets flux)
$\log_{10} P_{\text{total}}$ (Basin inhabitants)